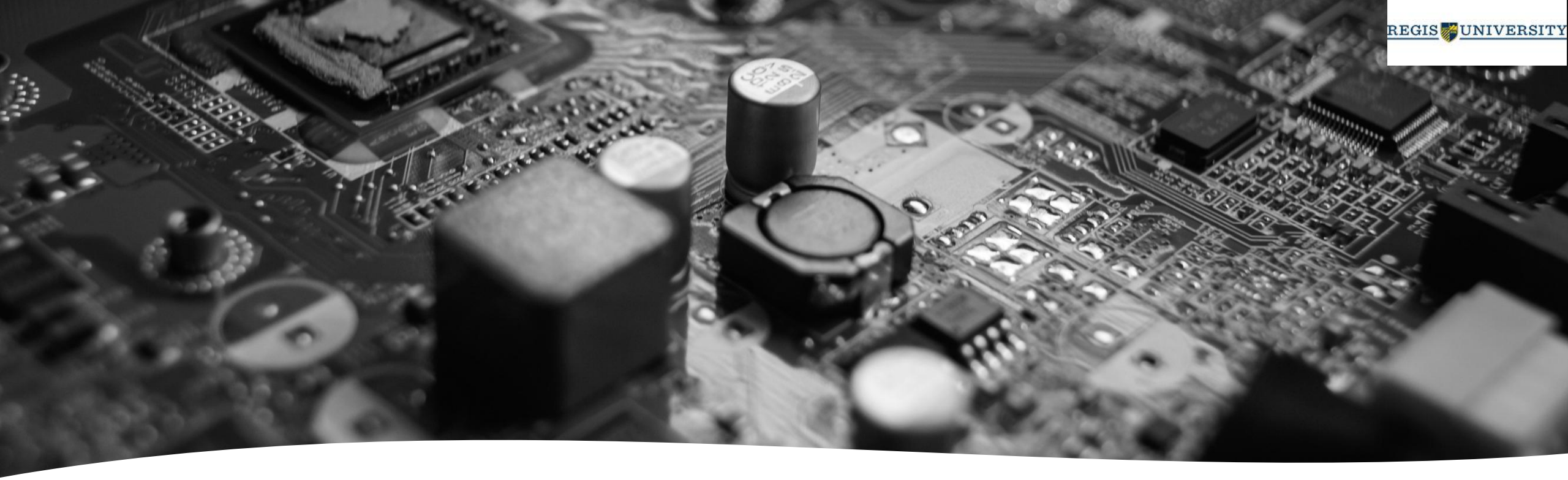

# Energy Implications of Prompt Design in Edge-Based LLM Inference


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MSDS 696 – Data Science Practicum II

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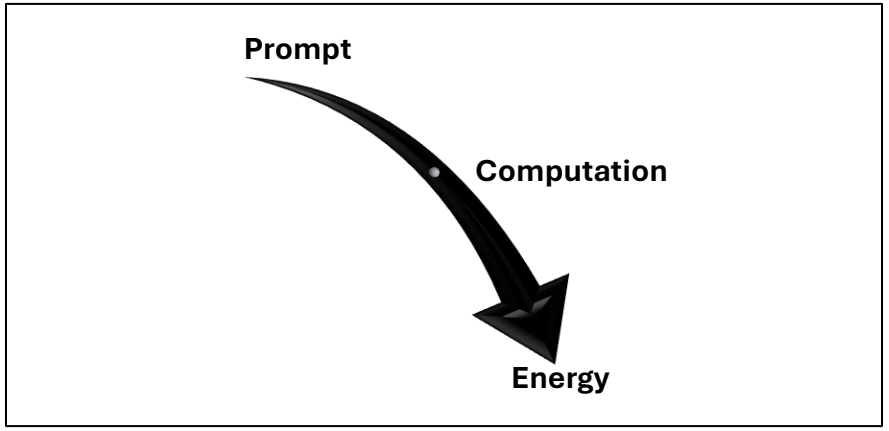
# 2. Problem & Goal

- Moving from prompt design analysis on open and closed LLM's to energy measurement in edge devices.
- Large Language Models (LLMs) require high computational resources.
- Edge devices (e.g., Jetson Orin Nano) have limited power capacity.
- Energy efficiency is critical for real-world deployment.




***Does Prompt Design Affect Energy Consumption during LLM inference?***

**Goal: Analyze how prompt styles affect tokens, latency and energy.**

# 3. Dataset & Experimental Setup

**3 Datasets**

- REPLIQA (2024-25) → Q/A
- MultiHopQA (2025) → Reasoning
- CCSum (2024) → Summarization

**Inputs**

- 20 input instances per dataset.
- Each instance represents a single task (question or document).
- 4 Prompt Styles
  - ❑Concise
  - ❑Structured
  - ❑Verbose
  - ❑Chain-of-Thought

**Total Runs**

- 60 Inputs x 4 Styles = 240 Runs

**3 Datasets** → **60 Input Instances** → **4 Styles** → **240 Runs**

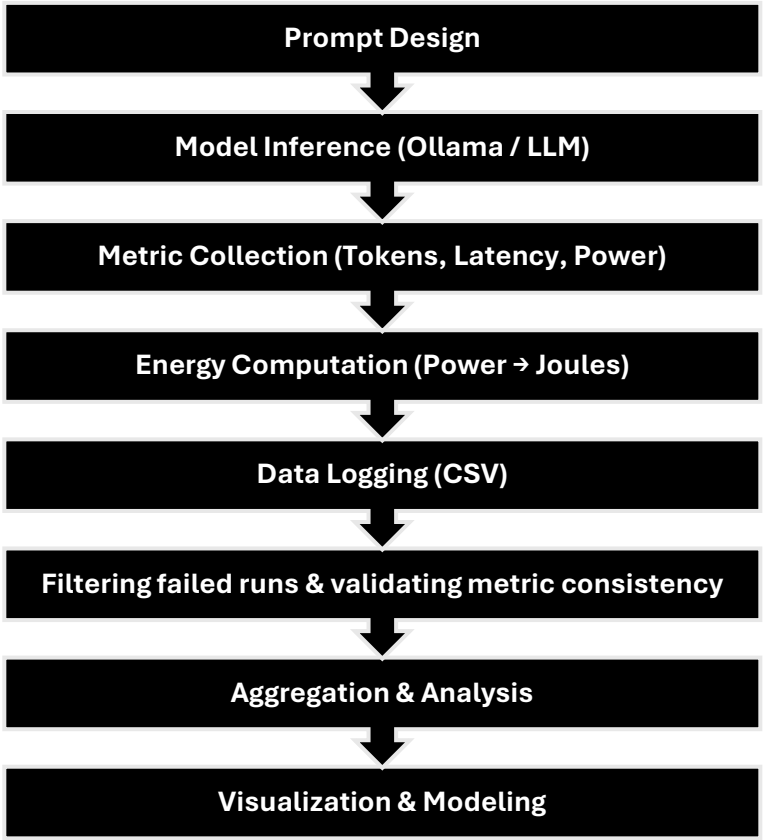
Device: **Jetson Orin Nano**

NVIDIA

Metrics Collected: **Tokens, Latency, Energy, Power**

# 4. Data Collection & Experimental Pipeline

**Built an automated pipeline that controls prompts, execution and metric logging across runs.**




### Experiments automatically iterate across:

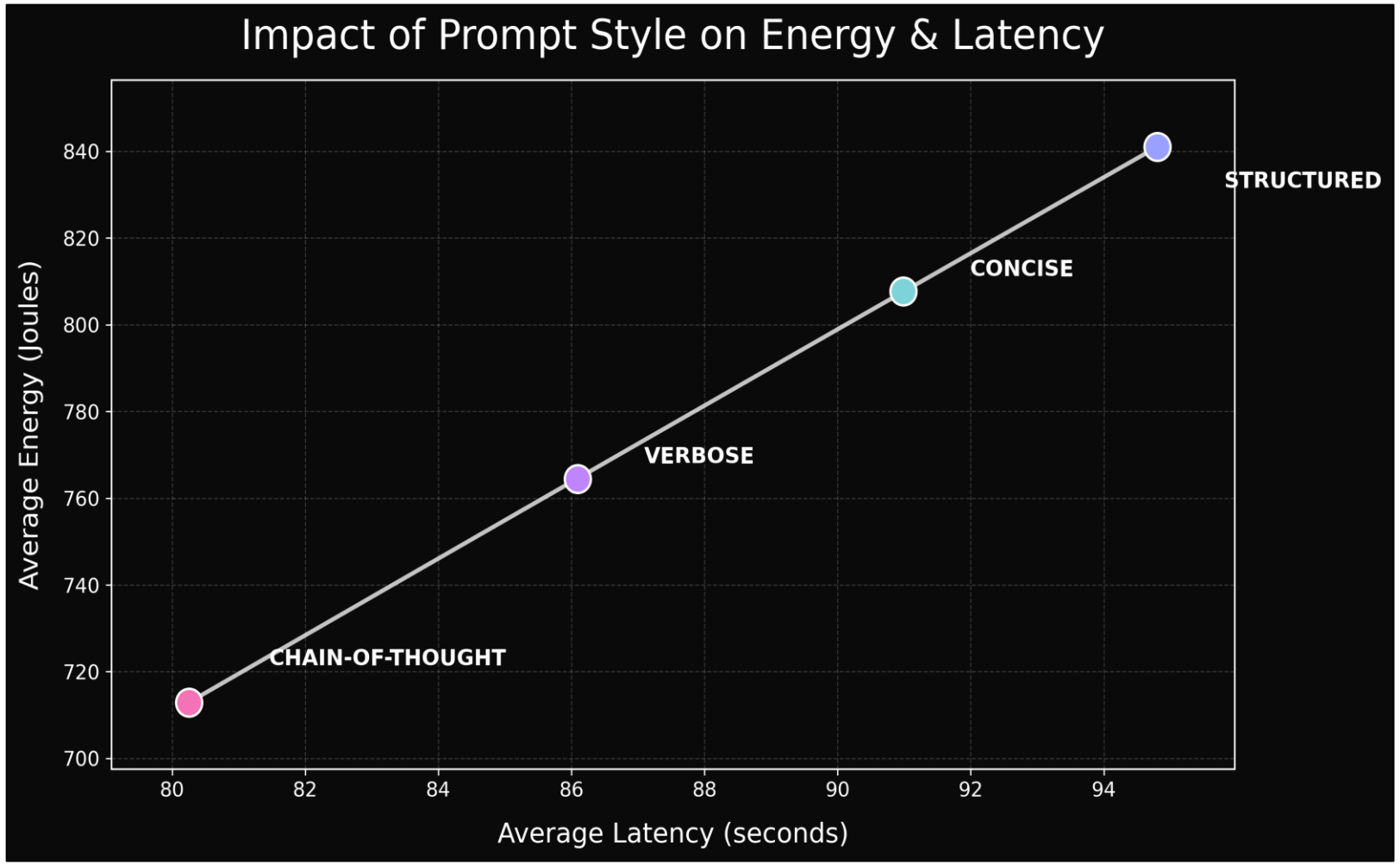
- Datasets
- Prompt Styles
- Each Configuration executed ~60 times for statistical reliability.

### For each run, we record:

- Input Tokens, Output Tokens
- Latency
- Energy & Power (via tegrastats)

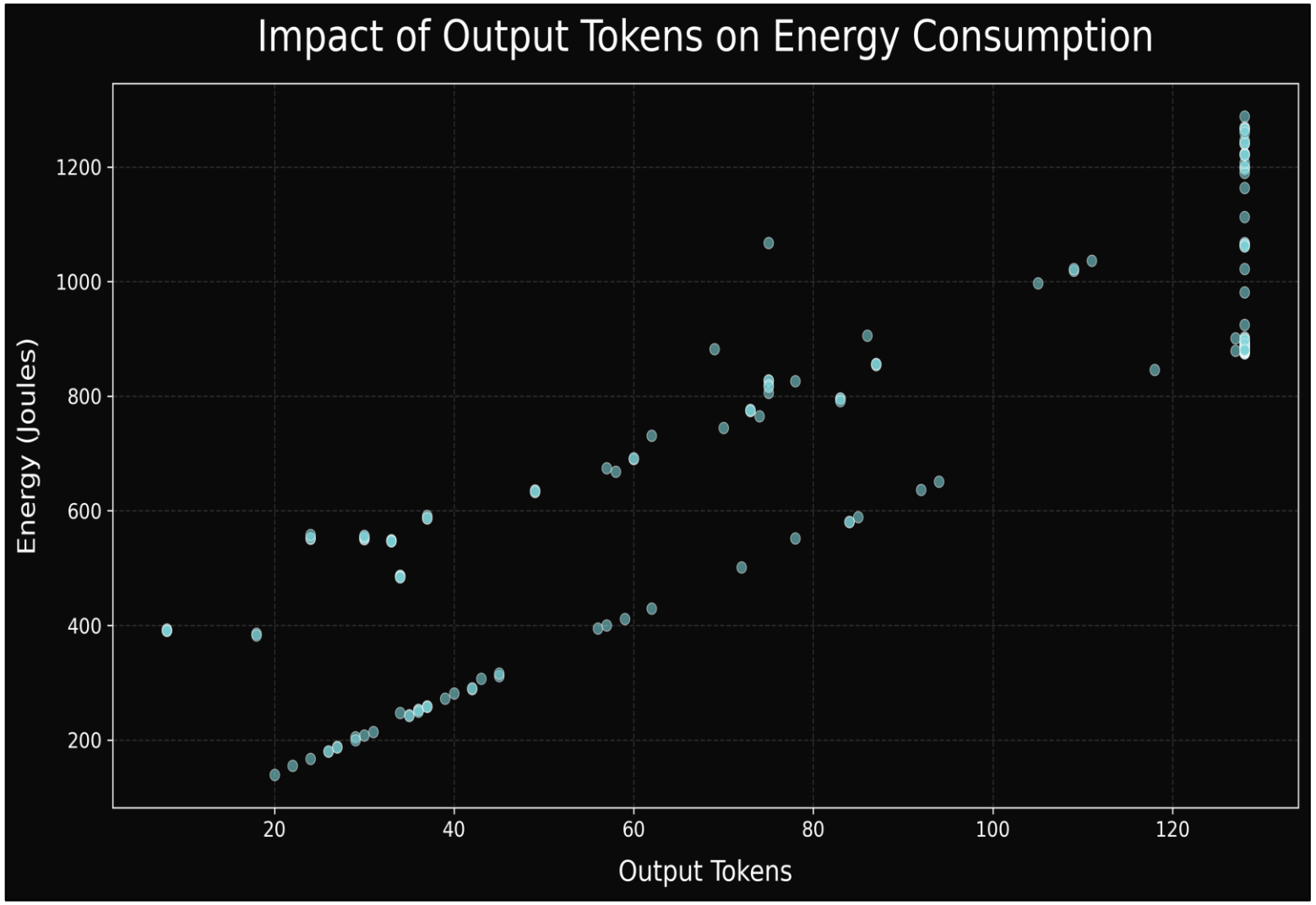
# 5. Impact of Prompt Style on Energy & Latency

- Energy consumption scales linearly with latency across all prompt styles.
- Structured prompts shows the highest consumption of energy followed by concise prompts.
- Structured uses ~18% more energy than Chain-of-Thought.
- Chain-of-thought prompts were unexpectedly more efficient in this setup.
- Each point represents average across 60 runs per prompt style.



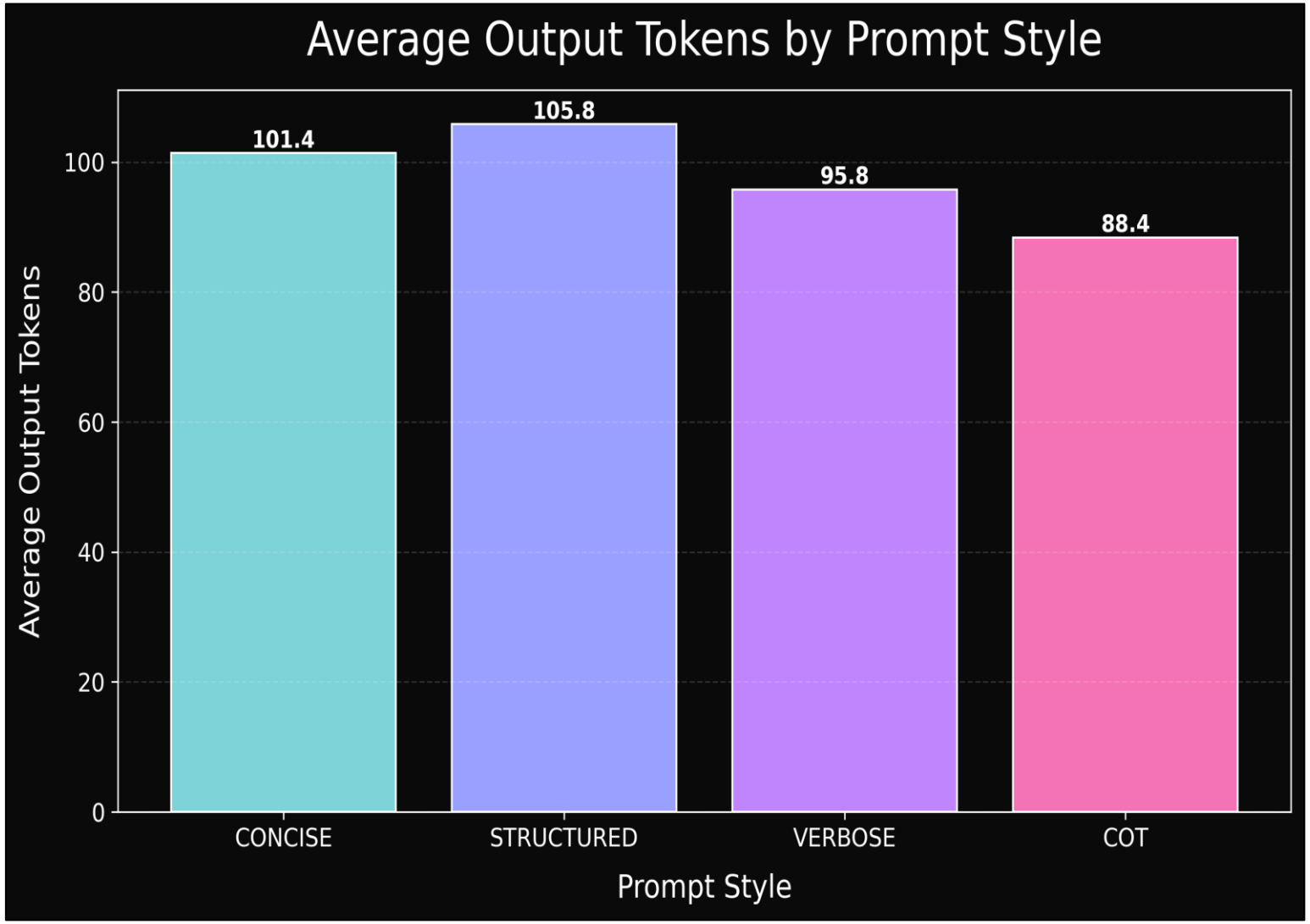
# 6. Impact of Output Tokens on Energy Consumption

- Energy consumption increases with output tokens length.
- Strong positive correlation (r ≈ 0.9).
- Longer responses require more computation and latency.
- Output tokens are the dominant driver of energy usage.
- Each point represents an individual run out of a total of 240 runs.



# 7. Average Output Tokens by Prompt Style

- The highest number of output tokens are generated by structured type of prompts.
- Chain-of-Thought prompts are the lowest.
- Energy variation can be observed by seeing the difference between output length among different prompt styles.



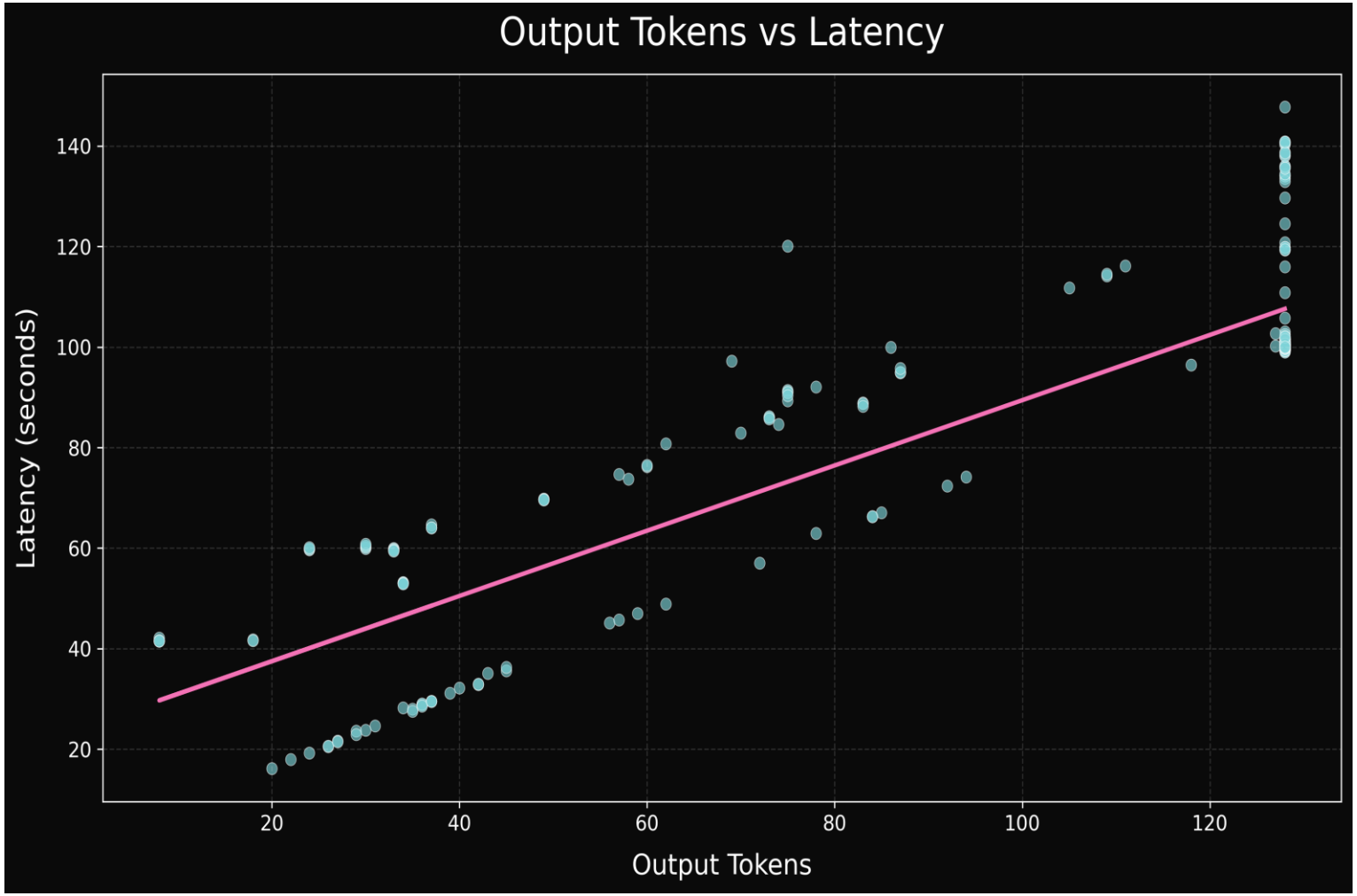
# 8. Energy Prediction: Pre vs Post Inference

- Pre-run models show weak predictive power ($R^2 \approx 0.13$).
- Post-run models achieve near-perfect accuracy ($R^2 \approx 0.999$).
- Runtime behavior (tokens & latency) determines energy.
- Runtime features dominate prediction.

| Model Type | $R^2$ | Meaning |
|---|---|---|
| Pre-run | 0.12 | Weak predictive power |
| Post-run | 0.999 | Near-perfect prediction |

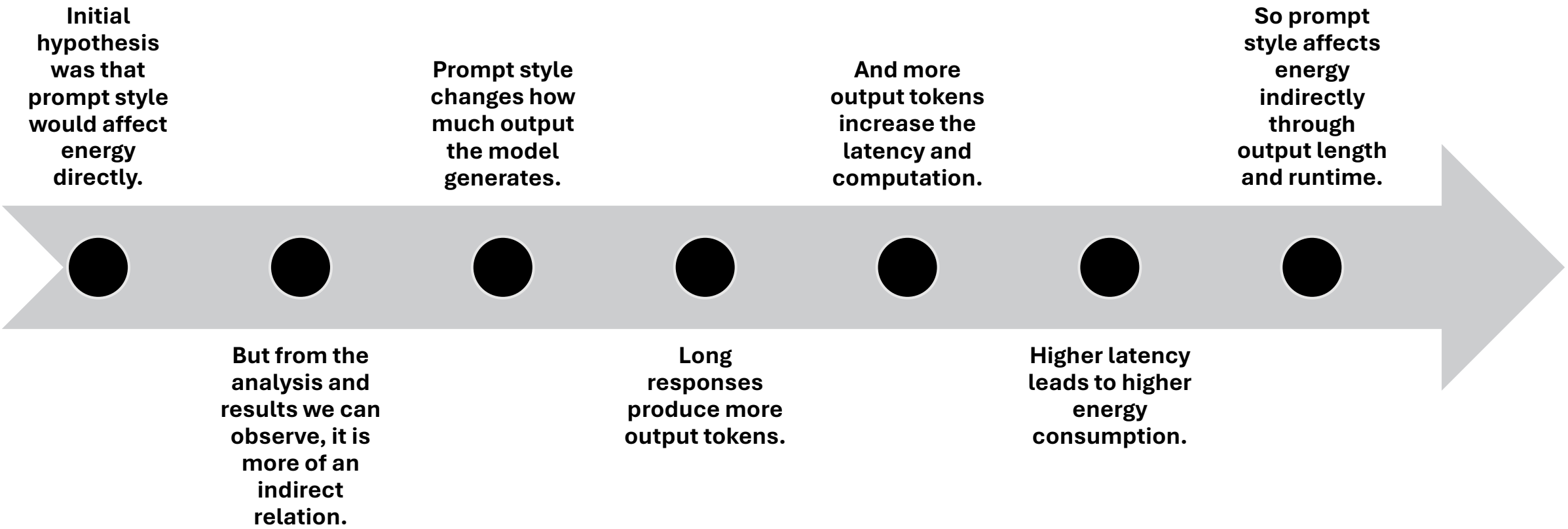
# 9. Regression Insights

- From the regression analysis we can observe long outputs increase latency.
- **Latency**, is the strongest predictor of energy usage.
- This concludes runtime behavior, rather than prompt style alone, is the main factor for energy consumption.




| Feature | Impact on Energy |
|---|---|
| latency_sec | Highest |
| output_tokens | High |
| Input_tokens | Low |

# 10. Key Insight: How Prompt Design Impacts Energy



# 11. Conclusion

Prompt design doesn’t directly affect energy

The main reason for energy use is output tokens.

Latency has a strong relationship with energy use.

Runtime behavior determines energy.

Efficient prompts can help improve edge AI sustainability indirectly.

# 12. Future Work and Improvements

- Improve pre-inference energy prediction models.
- Test across additional Jetson power modes and clock speeds.
- Analyze trade-offs between response quality and energy consumption.
- Develop adaptive prompting strategies based on resource constraints.

# 13. References


1. E. J. Husom et al., “Sustainable LLM inference for edge AI: Evaluating quantized LLMs for energy efficiency, output accuracy, and inference latency,” arXiv, 2025.
2. M. Arya and Y. Simmhan, “Understanding the performance and power of LLM inferencing on edge accelerators,” arXiv, 2025.
3. E. J. Husom et al., “The price of prompting: Profiling energy use in large language models inference,” arXiv, 2024.
4. U. E. Dammu, “Energy efficiency in LLM inference: Open vs. closed models,” Regis University Practicum I, 2025.
5. ServiceNow, “REPLIQA dataset,” Hugging Face, 2024–2025.
6. corag, “MultiHopQA (HotpotQA configuration),” Hugging Face, 2025.
7. CCSum Team, “CCSum summarization dataset,” Hugging Face, 2024.
8. NVIDIA Corporation, “Jetson Nano Developer Kit Technical Reference Manual,” 2023.
9. F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” JMLR, 2011.
10. TinyLlama Team, “TinyLlama-1.1B-Chat-v1.0,” Hugging Face Models, 2024.
11. OpenAI, “GPT-5 Pro and GPT-4.1 Mini large language models,” 2025. [Online]. Available: https://platform.openai.com

# Thank you

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https://github.com/emmaniuel26/Practicum-2